

Turpan, Xinjiang, Northwestern China

Karez Technology for Drought Disaster Reduction in China

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Abstract

Karez is a traditional irrigation water system which is able to make use of underground water efficiently. The Karez system has a long history in the Xinjiang area of China. As a comprehensive system, Karez is composed of four primary components: vertical wells, underground canals, a surface canal and small reservoirs. Because of the Karez system, Turpan, a basin located in the arid area of Northwestern China, is well-known for its wide variety of agricultural products. In the Turpan area of Xinjiang, Karez is still being used to supply water resources for irrigation and domestic uses. At present, modern technology has been integrated into the traditional Karez system to further reinforce the successful traditional practice.

Background

The Turpan depression, with a height of 32.8m, is located in the Turpan basin which is the second lowest basin in the world. It is surrounded by several high mountainous areas (3500~5000m) which are covered with glacier or permanent snow. The minimum height of Aiding Lake, at the south of the basin, is around -155m, which represents the lowest lake in China. Underground water is abundant in the shallow underground water-bearing layer. The terrain of the mountains around Turpan basin is mainly formed by Hercynian movement at the end of the Paleozoic period. It is hard and fractured and therefore fissured water is easily formed. The exposed rocks of Flaming Mountains mainly consist of sandy conglomerate and mud stones of Jurassic, Cretaceous as well as Tertiary. Therefore, the geological condition of Turpan district is fit to construct an underground canal with little reinforcement for amassing sufficient water resources. The Turpan area is famous for its wide variety of fruits such as grapes, watermelons, and Hami muskmelons.

Story/Event

Turpan is very dry in all seasons and very hot during spring, summer and autumn. The highest temperature recorded is 47.7°C in summer. High temperature and strong solar radiation result in high annual evaporation amounting to 2800~3000 mm.¹ Turpan is in an inner land with total annual precipitation of only around 16~17 mm. Because of strong evaporation or the evapotranspiration process, precipitation (rain or snow) falling on the slope of the mountain would evaporate or seep underneath sand and soil before it can converge into a stream and reach the flat agricultural area along the foot of the mountain. Surface water is scarce in most areas. Under such tough circumstances, few plants or animals survive.

¹ Ji ZHAO (2001).

Indigenous Knowledge

Karez is a traditional irrigation water system with a long history in Xinjiang area of China which makes use of underground water efficiently. Where farmland is located in the mountainous area, it is built on an alluvial fan or plain. Most existing Karez systems were mainly constructed between the 17th and 20th century. The currently functioning Karez systems are distributed in the dry areas of the southern slope of Mountain Tianshan in eastern Xinjiang, in the Hami district as well as the Turpan, Shanshan and Toksun districts of Turpan Basin. In the Turpan district, 1,016 Karez systems still exist of which 686 are operational. The total length is about 3,000 kilometers. The average depth of the underground canal is 20 meters whilst the utmost depth is 90 meters. The total outflow of the karez systems in Turpan Basin is about 10 cubic meters per second accounting for about 20% of the total diversion water of the basin.² At present, modern facilities such as electromechanical wells are integrated into the system.

The structure of a complete Karez system can be complex but its basic structure is essentially composed of four main components: vertical wells, underground canals, surface canal, and small reservoirs (Figure 1).

Vertical Well

The length of the underground canal varies from around 3 km to 30 km. It is almost impossible to dig such a long underground canal without digging a vertical well, especially in the past agricultural period

when no modern equipment was available. Hence, the vertical well is primarily used to assist in digging the underground canals. When the wells are dug, sand and soil are excavated with the help of animals.

The major functions of vertical wells are for ventilation, proper orientation of the canal during construction and examination and repair of canals after construction. The distance between vertical wells is generally around 60~100 meters in the upper reaches, 30~60 meters in the middle, and 10~30 meters in the lower reaches. The depth of wells is approximately 40~70 meters, 100 meters in the upper reaches, 30~40 meters in the middle, and 3~15 meters in the lower reaches. Vertical wells are utilized not only to aid in digging underground canals, but also to take out water from these canals after the whole Karez system is completed. An aerial image of vertical wells is presented in Figure 2.

Underground and Surface Canals

Of the two types of canals, the majority of them are underground. The canals under the surface are generally part of a network which enables underground water to accumulate (as illustrated in Figure 3a). Usually the deep soil around the underground canal is very strong and does not collapse easily. However, when the canal is close to the surface, the soil is loose. To prevent the underground canal from collapsing, it is usually reinforced with wooden pillars. At the bottom of the well, both sides are excavated for the underground canal. When the underground canal reaches the farmland, it becomes a surface canal and is linked to a small reservoir or directly connected to a system of water channels for

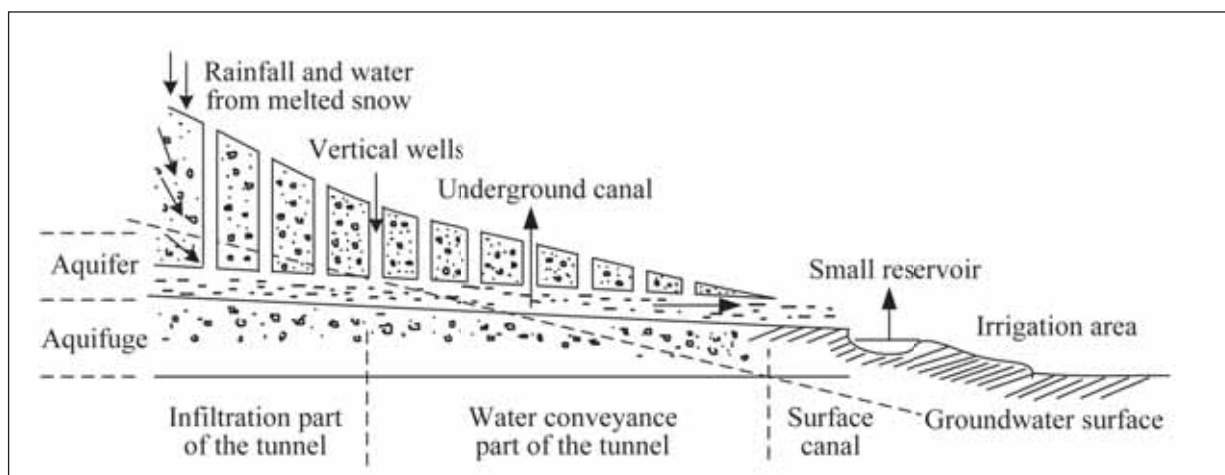


Figure 1.
The components of a typical Karez system

irrigation (Figure 3b). Usually the surface canal is short to limit evaporation.

Small Reservoir

Water resources are collected in small reservoirs which can be adjusted for water level and temperature. Constructing reservoirs increase the water level so that a larger area of farmland is irrigated. Moreover, the water stored in reservoirs receives sunlight which increases their temperature. The warmer water is more appropriate for irrigation, since the low temperature of water due to snow melting or because of its underground source may do severe harm to crops.

A variety of simple tools are employed for building Karez systems. These include an excavating hoe, a planning hammer, a basket, a winch and an oil lamp (as displayed in Figure 4). The excavating hoe and planning hammer are used for digging passages underground. The basket and winch are used for carrying soil and sand. Iron oil lamps with an arrow for orientation are used for digging underground canals. The lamp can also be conveniently fixed on canal walls. Currently in Hami city of China, a sunlight reflecting mirror is also used for orientation.

In constructing the different components of the Karez system, it is vital to ensure that along the underground canal, water resources are sufficient. A crucial first step is to find water sources in the upper reaches and the depth of water estimated in accordance with the location of farmland. The location for digging the well can then be decided. Subsequently, wells and canals can be built gradually from the lower to the upper reaches following the water source.



Figure 2.
Aerial image of vertical wells in Turpan area of China. Source: www.sunnychina.com



Figure 3.
(a) underground canal reinforced with wooden pillars. Source: www.chinahw.net ; and
(b) outreach of underground canal to surface canal. Source: www.travelchinaguide.com

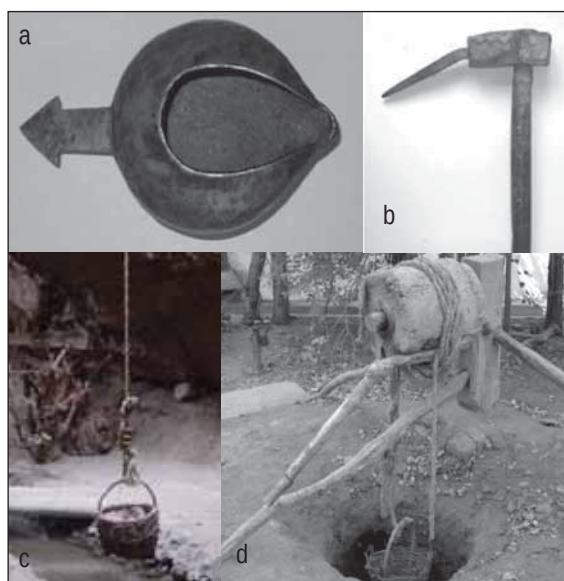


Figure 4.
(a) traditional oil lamp with side-arrow, (b) excavating hoe, (c) basket
and (d) a modern sheave used in building the Karez system.
Source: www.karez.org, and cersp.com

Lessons Learned

The Karez system is a proven and effective indigenous drought reduction technology that is still in use. This traditional knowledge has several advantages as follows:

1. **Support by Earth's Gravity.** Since Karez takes advantage of topography to divert deep subsurface flow through an underground canal to land surface for gravity irrigation, the cost for water-lifting equipment and its maintenance are almost negligible.
2. **Stable Outflow.** The major water sources of the Karez system are melting snow and underground water. The underground canal can minimize high evaporation in the windy Turpan district hence, the impact of climate change is small. In addition, problems with sand blast can be avoided in the underground canal. All this makes the Karez system able to provide stable water resources, though total water volume may not be very large. As observed, there has been a very stable population for thousands of years in Karez areas, regardless of environment changes.
3. **High Water Quality.** Melting water from snow enters the system and the soil provides a very good filter to remove polluted materials. Unlike water channels on land, the underground canal minimizes water pollution and at the same time is rich in minerals. The water quality is suitable for drinking and domestic use.
4. **Construction with Simple Tools.** Most Karez systems are built with simple tools and do not require complex equipment.

Conversely, the Karez system has constraints. There is a spatial limitation on its construction and use. It is only applicable in areas with stable underground water supply and hard soil. Some Karez systems were built around Guanzhong Plain in Central China in the Han Dynasty which did not last because the underground canals collapsed. Furthermore, the volume of water in the Karez system may change with the season although there is little variation in the volume daily. In the summer, there are sufficient water resources if the water source is from melting snow. In the spring, Karez water is limited, while in autumn and winter it is substantial. This is sometimes in discord with water demand for agriculture.

There is thus a need to strengthen the traditional Karez system with modern technology. Its value as an efficient indigenous drought reduction technology should not be ignored. Instead, this traditional knowledge should be improved and reinforced with modern technology. Its widespread use promoted in the face of severe drought disaster in the future.

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An Hai Commune, Ninh Phuoc District, Ninh Thuan Province, Vietnam

Weather Forecasting through Indigenous Knowledge for Crop Cultivation in the Drought Prone Area of Vietnam

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Abstract

The coastal area of Ninh Thuan province is the driest area of Vietnam. The annual rainfall is normally between 600 mm and 800 mm per year with uneven distribution throughout the month. Since weather forecasting is still limited in some low developed areas, the use of indigenous knowledge for weather forecasting is very useful in crop cultivation in the community. Specifically, the methods of moon observation and dragonfly observation for predicting weather changes have played an important role in agriculture activities in the An Hai community.

*“Dragonfly flies high, sunny sky
Flies low, rain
Flies neither high nor low, cloudy sky”*

(Lyrics of a folk song used in weather forecasting)

Background

An Hai is one of several coastal communes located to the East of Ninh Thuan province. The total geographical area is 2,091.98 ha with 1,057.37 ha in agriculture, 4239 ha in forestry, 4833 ha in dwelling and the rest in aquaculture and other uses. In 2005, the total population of the An Hai commune was 12,890, divided into about 2,596 households in a total of 6 hamlets.

The main livelihood of the community is agriculture, with rice, grapes and short-duration vegetables (such as tomatoes, carrot, chili and potatoes) as main crops. Animal husbandry is another one of the strengths of the An Hai commune, focused mostly on raising cattle, sheep and goat.

There are three rice crop seasons in irrigated areas: Winter-Spring from early Dec to early April; Summer-Autumn from late April to early August and main cropping season from September to early December.

The primary sources of water in the dry season are river water and well water. In the rainy season, people mainly use rainwater for crop cultivation.

Story/Event

Temperatures in inland Vietnam are rising and the weather is becoming extreme and unpredictable. Average temperatures are now 1° C higher than they were about 100 years ago. Changes in rainfall patterns are complex and vary depending on the location; however, the main trend is towards hotter, longer and more arid dry seasons and more intense rain in the wet season.

Ninh Thuan is one of the nine provinces that are most affected by drought in Vietnam. This province has experienced yearly drought since 2002. There are two seasons in this area: the rainy season from July to November and the dry season from December to June. Normally, the amount of rainfall received in the coastal regions is very low. The annual rainfall in the coastal town of Phan Rang – Thap Cham (very near the An Hai commune) is 712 mm. The rainy season is heavy during the three months of September to December.

The province was severely affected by drought in August 2004 with a 50% reduction in rainfall compared to the

norm. The drought continued in 2005 and 2006 with poor rainfall during the first two cropping seasons. Prolonged dry days have caused significant damage to agriculture and changed the salinity of groundwater thus damaging aquaculture. The increasing drought events in the Ninh Thuan province are major concerns for both government and local communities.

Whilst annual rainfall has been steadily increasing, the farmers experience droughts because rain now comes in intense, concentrated bursts. They have to change their crops and crop schedule to adapt to the hard weather and climate conditions. Communities are always seeking new ways to adapt. Farmers work together to grow crops and keep herds of goats and sheep of the Sultan breed from India, which can tolerate high temperatures. Farmers have switched from growing rice to maize or other short-time seeds and changed to more drought-resistant varieties. They have devised ways to economize on water use and reuse water. In particularly difficult times, family members migrate to the cities to find work. Some of these adaptation techniques are productive, but others carry a cost. Migration of young people increases the workload of the older people left behind. Women sometimes go without water in order to give it to their husbands or children.

Indigenous Knowledge

Through weather forecasting, recommendations on planting dates and changing agricultural varieties can be given to make sure that rice and other crops will grow. Since weather forecasting systems were not available in the past, many farmers cultivated their crops based on moon observation and by watching the habits of insects.

Moon Observation

The An Hai community has an old proverb that has been used in weather observation for drought forecasting and crop scheduling. The proverb is passed on to generations in the whole community. The proverb states:

“Trăng quầng thì hạn, Trăng tán thì mưa”

In English, it means,
“Corona around the moon, there will be a drought year;
Halo around the moon, rain soon.”

Corona Moon

A corona is a type of plasma “atmosphere” of the sun or other celestial body, extending millions of kilometers into space, most easily seen during a total solar eclipse, but also observable in a coronagraph. The Latin root of the word corona means crown.

The high temperature of the corona gives its unusual spectral features, which led some to suggest in the 19th century that it contained a previously unknown element, “Coronium”. These spectral features have since been traced to highly ionized Iron (Fe (XIV)) which indicates a plasma temperature in excess of 10° Kelvin (Aschwanden, 2004).

Halo Moon

A halo moon is a ring of light that surrounds either the sun or moon and typically appears as bright white rings.

However, some halos can also have color patterns. A halo is an optical phenomenon similar in concept to a rainbow but also very different. Halos form when the sunlight or moonlight is refracted or bent by ice crystals associated with thin-high level clouds, like cirrus or cirrostratus. The most common type of halo is the 22 degree halo. In this halo, a ring of light 22 degrees from the sun or moon is projected by hexagonal (i.e. six-side) ice crystals with diameters less than 20.5 micrometers.¹ Often, a halo around the moon or sun is an indicator of cloudy or rainy weather as high level cirrus and cirrostratus clouds that cause halos tend to drift ahead of frontal systems (especially warm fronts) that produce rainfall.

Dragonfly Observation

The An Hai community uses another folk song for weather forecasting. They observe the habits of the dragonfly to know when it will rain and when there will be sunshine. The Vietnamese lyric of the folk song is as follows:

*“Chuồn Chuồn bay thấp thì mưa,
Bay cao thì nắng,
Bay vừa thì râm”*

In English, it is translated:
“Dragonfly flies high, sunny sky
Flies low, rain
Flies neither high nor low, cloudy sky”

The farmers explained that when dragonflies fly a distance lower than 80 cm from the ground, it will rain soon. Based on that, the farmers prepare soil and seeds for the crops. Using the method of dragonfly observation, the farmers can decide the time to sow and plant, and make their crop schedule appropriately.



Figure 1. Corona Moon
Source: http://www8.ttvnol.com/forum/f_533/862476.ttvn

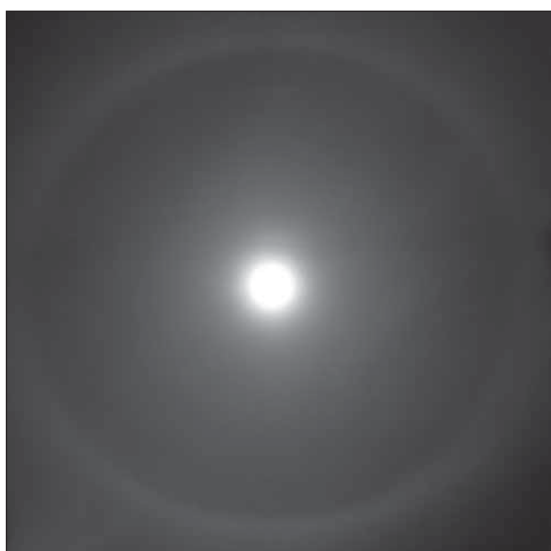


Figure 2. Halo Moon
Source: http://www8.ttvnol.com/forum/f_533/862476.ttvn

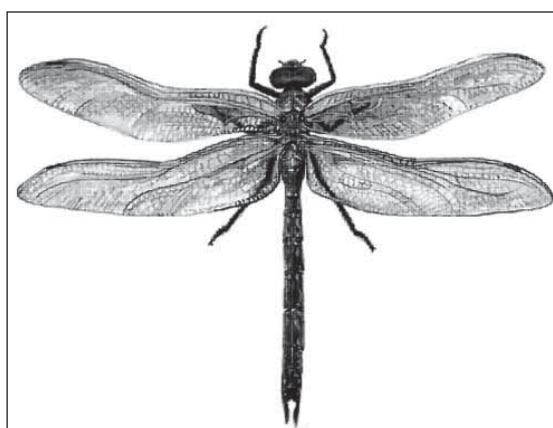


Figure 3. A dragonfly
Source: <http://www.scientificillustrator.com/illustration/insect>

¹ NASA (2003)

Lessons Learned

Observation of insects and atmospheric phenomena such as halos were used as an empirical means of weather forecasting before meteorology was developed. The indigenous knowledge is passed on through thousands of years and from generation to generation by the community. These are based on practical experience and have been in use for a long time. In some regions where the residents do not have access to high technology, traditional weather forecasting based on observation of the moon and insects play an important role in agriculture activities.

The main problem in the An Hai community is lack of fresh water for crop cultivation. Knowledge of the rainy season and forecast of the exact drought year helps farmers choose crops, seeds and the time to sow and plant. Since this traditional knowledge is very useful in regions which do not have access to meteorological methods of forecasting, these strategies can and should be disseminated to other communities in low developed regions.

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